Innovation Video Script

Intro / Problem:
Space feels empty.

But even in low Earth orbit, it isn't a vacuum.

At just five hundred kilometers above Earth, thousands of satellites still skim a thin layer of atmosphere, enough for drag to matter over time.

Left alone, that drag will eventually pull satellites back toward Earth and burns up. But without intervention, that process can take decades.

During those decades, inactive satellites become long-lived debris, increasing collision risk, forcing costly avoidance maneuvers, and threatening the sustainability of space itself.

That's why international guidelines require satellites to deorbit within twenty-five years [on screen: ADD NOTE NEW REGULATION 5 YEARS] of mission end. Yet according to the European Space Agency, compliance remains insufficient.

So the problem isn't awareness.
It's implementation.

Current Market:
Many current deorbit solutions, like traditional drag sails, rely on fragile materials, complex hinges, or moving parts that can jam, fail, or consume valuable power.

Others are too bulky, too expensive, or too unreliable to scale across the rapidly growing small-satellite market.

So the question becomes:

Could deorbiting be simpler, smaller, and more reliable by design?

AeroFold Intro:
We are AeroFold: [insert tagline].

AeroFold is an origami-inspired, tessellated deorbit module that fits inside a fraction of a CubeSat unit and unfolds into a large, rigid drag sail at end-of-life, bringing satellites back to Earth in a year instead of decades.

It's compact.
It's robust.
And it integrates seamlessly into existing satellite systems.

But how does it work?

<u>AeroFold Mechanics: [FINISH/FIXX]</u>
At the core of AeroFold is a square origami tessellation manufactured from nitinol, a shape-memory alloy. Unlike conventional drag sails that rely on hinges or bearings, AeroFold has no traditional moving parts. Instead, the nitinol structure is trained to remember a precise deployed shape.

During launch, the module is folded into a compact box configuration, making it easy to integrate into almost any satellite platform.

At end-of-life, a small electric current gently heats the nitinol.That heat triggers the material to return to its programmed geometry, unfolding the tessellated sail in a single, controlled motion. As a result, our design expands to nearly 225 times its original cross-sectional area, which means drag force and orbital deceleration is multiplied by a factor of ~225.

Even if the structure is slightly damaged during launch or in orbit, the material self-corrects, always returning to its intended shape.

One of the greatest aspects of the AeroFold design is that each of the 253 rigid segments can be individually controlled, allowing the design to change its configuration actively. Since the satellite can now finely adjust its attitude, it won't just deorbit, it'll deorbit in a controlled and predictable manner.

The result is a deployment system that is passive, reliable, and mechanically simple, no matter what space throws at it.

<u>Validation & Tests [FINISHH/FIXX]:</u>
To validate our design, we used multiple simulation tools, including Autodesk Inventor, to convert different tessellation geometries from 2D to 3D. Our goal was to maximize surface area while minimizing volume and mechanical complexity.

Through this process, we identified square tessellations as the optimal configuration. The square tessellation would fold into a box-like shape, making it easy to implement in a satellite which usually already has rectangular features. Additionally the simplicity of each segment's shape allows simpler manufacturing, which would reduce cost.

We then modeled aerodynamic performance using OpenFOAM, mapping drag coefficients across orbital conditions. In this simulation, we attached AeroFold to a replica of the SpaceX Starlink V2 satellite, which is one if not the most common satellite design in orbit currently, and it was found that the satellite would successfully deorbit in less than five years after the dragsail was deployed.

We have already patented this design , finished the conceptual design, and numerical simulations. Our next step is to build the first deployable prototype for lab testing.

<u>Competitive Comparison:</u>

Based on our numerical simulation and data, we compared our product to our competitors. AeroFold has the same compliance as ADEO/Spinnaker, except at half the price, mass, and volume with no moving parts and a 225x expansion ratio.

<u>How a Real Mission Uses AeroFold</u>

AeroFold is designed to fit directly into the workflow satellite operators already use.

For a 3U Earth-observation CubeSat, the mission planner simply enters mass and orbital parameters into our configurator and selects an AeroFold unit to meet the desired deorbit timeframe.

During integration, the module bolts directly onto the satellite and connects to the onboard computer using standard interfaces.

At end-of-life, a single command deploys the sail. The team receives immediate confirmation that deorbiting has begun, no additional steps, no uncertainty.

<u>Conclusion:</u>

Space feels empty.

But every inactive satellite left behind makes it more crowded, more dangerous, and more fragile.

As access to space expands, sustainability can no longer be an afterthought. it has to be designed into every mission from the start.

AeroFold transforms deorbiting from a complex, failure-prone process into something simple, reliable, and automatic. By using smart materials and origami-inspired engineering, we ensure satellites don't linger as debris for decades.

Instead, they do what they were always meant to do at the end of their mission:
reenter Earth's atmosphere and safely burn up.

Because the future of space exploration doesn't depend on how many satellites we launch.

It depends on how responsibly we clear the space we leave behind.